



Enhancing Mobile Cybersecurity Through Smishing Detection and Awareness Using Machine Learning Techniques

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Introduction

The Problem: Smishing (SMS phishing) attacks are rapidly increasing, exploiting mobile users through deceptive messages that bypass traditional security systems.

Current Limitations:

- Traditional systems rely on fixed rules, which are ineffective against sophisticated social engineering attacks
- Lack of interpretive mechanisms prevents users from learning to identify threats themselves
- Lack of Arabic language support

Our Solution: SmishGuard an intelligent web-based system that detects smishing using ML while teaching users to recognize threats through explainable visualizations. Trained on the SMS Phishing Dataset (Mendeley) and SMS Smishing Collection (Kaggle), the system combines DistilBERT classification with LIME explanations. Figure 1 demonstrates the system's real-time analysis with LIME explanation.

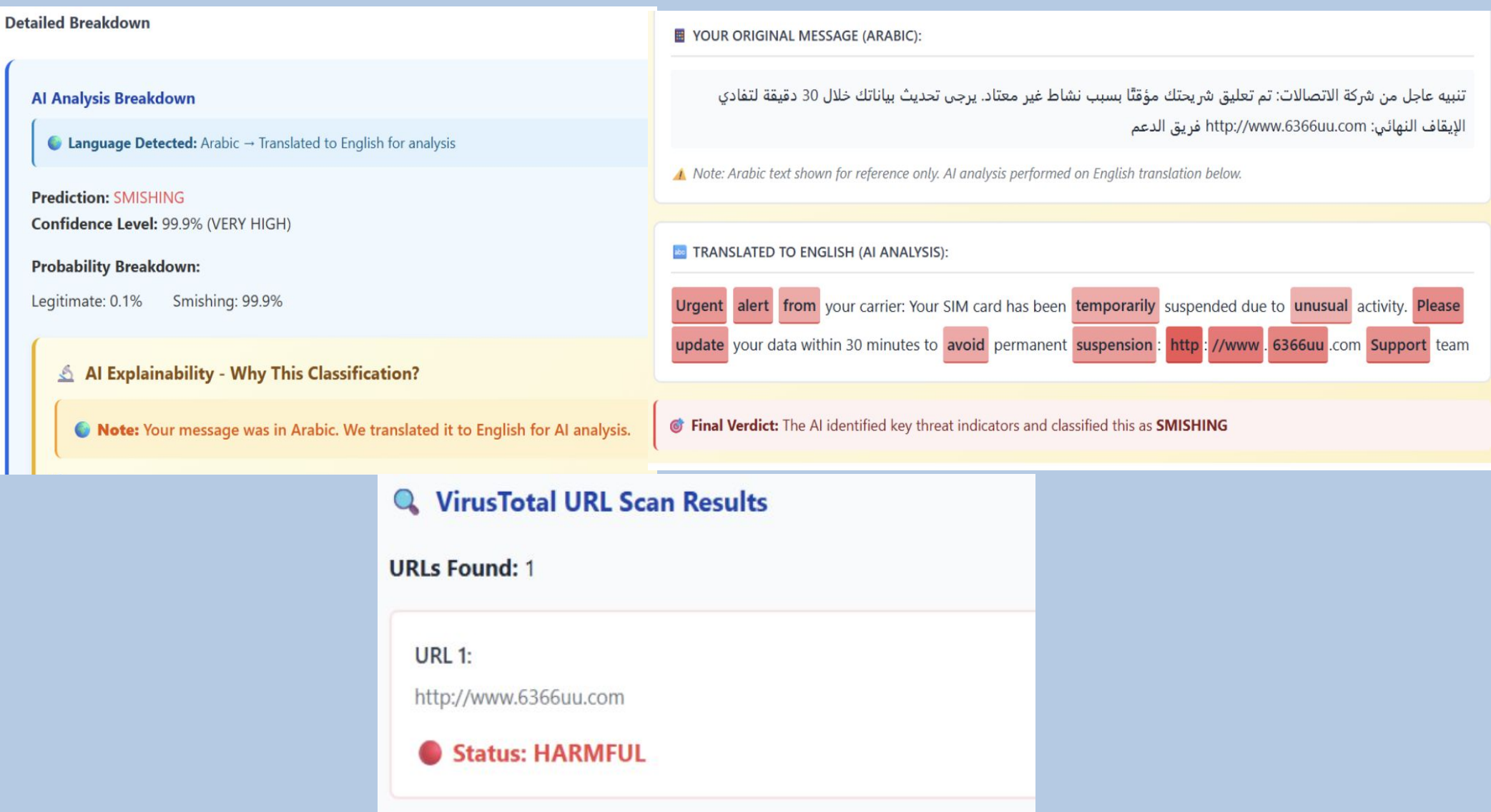


Figure 1: SmishGuard analyzing a smishing message with LIME explanations—red highlights indicate smishing threats, and green shows legitimate terms

Objectives

SmishGuard system aims to:

- Accurate Detection**
Comparative study of ML/DL models to identify the best smishing classifier
- Explainable AI**
Integrate LIME to show which words trigger detection
- Multilingual Support**
Enable Arabic and English message analysis
- URL Verification**
Integrate VirusTotal for real-time link scanning
- User Education**
Provide interactive quizzes and security guidance
- Web-based user interface**
Deploy accessible Flask-based application

Design

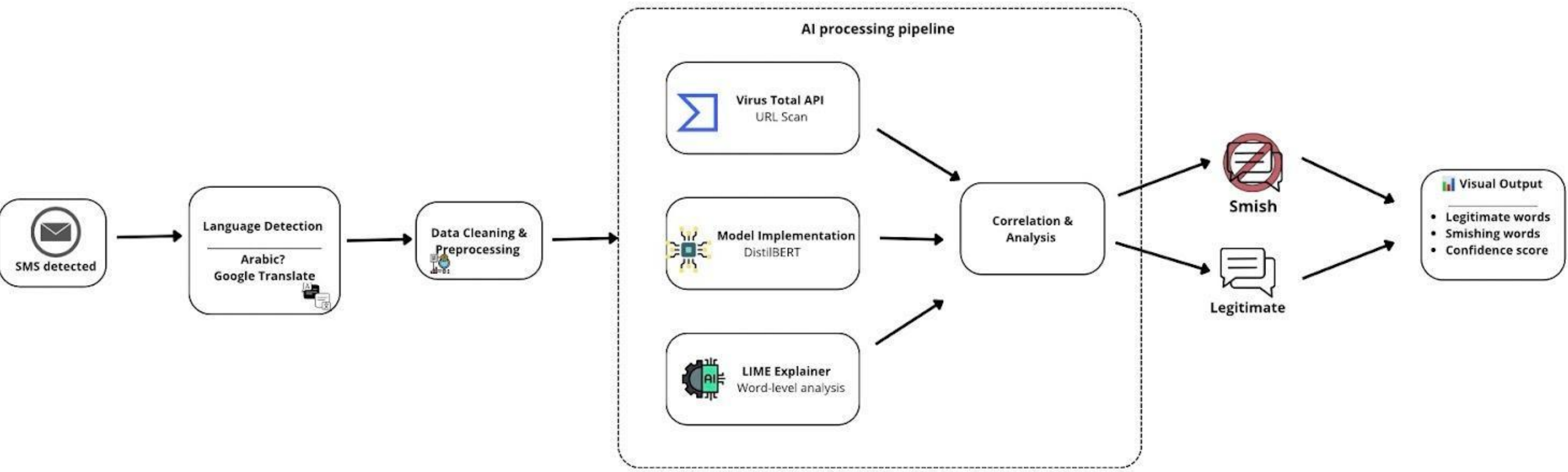


Figure 2: SmishGard Architecture Flow

System Architecture Overview

The proposed system analyzes SMS messages through a multi-stage intelligent pipeline. Messages are first language-detected and if it is in Arabic will be translated into English, then cleaned and preprocessed for robust analysis. Embedded URLs are extracted and scanned using the VirusTotal API to assess external threats. The processed text is classified using a fine-tuned DistilBERT model, while LIME provides word-level explanations for transparent decision-making. Final classification integrates both model predictions and VirusTotal results—prioritizing confirmed malicious URLs—and presents the outcome through a clear, user-friendly visual interface.

Model Development:

- Dataset Preparation:** Two SMS datasets were merged and cleaned.
- Augmentation Pipeline:** Applied to balance classes
- Final Dataset Size:** 8,007 SMS messages (Ham + Smish)
- Data Split:** Stratified split into train/validation/test (approximately 72%/8%/20%) to preserve class distribution
- Models Evaluated:** 6 ML and DL models were compared.
- Final Selection:** DistilBERT model (99.50% accuracy)

Results

Model Performance Comparison:

Table 1 shows performance metrics across six evaluated models,

Table 1: Comparative Performance of ML/DL Models—DistilBERT achieves superior balanced performance across all metrics

Model	Accuracy	Precision	Recall	F1-Score
RandomForest	0.9781	0.9877	0.8696	0.9249
XGBoost	0.9807	0.9763	0.8967	0.9348
CNN	0.9807	0.9305	0.9457	0.9380
LSTM	0.9798	0.9124	0.9620	0.9365
CNN-LSTM	0.9865	0.9468	0.9674	0.9570
DistilBERT	0.9950	0.9976	0.9835	0.9905

System Performance Metrics:

The DistilBERT model performed the best in all primary parameters, showcasing an accuracy of 99.50% and an F1-score of 99.05%. The precision level of the model at 99.76% shows that almost all messages identified as smishing were indeed harmful, while the recall value of 98.35% reflects its high capability to detect real smishing attempts with very few missed threats.

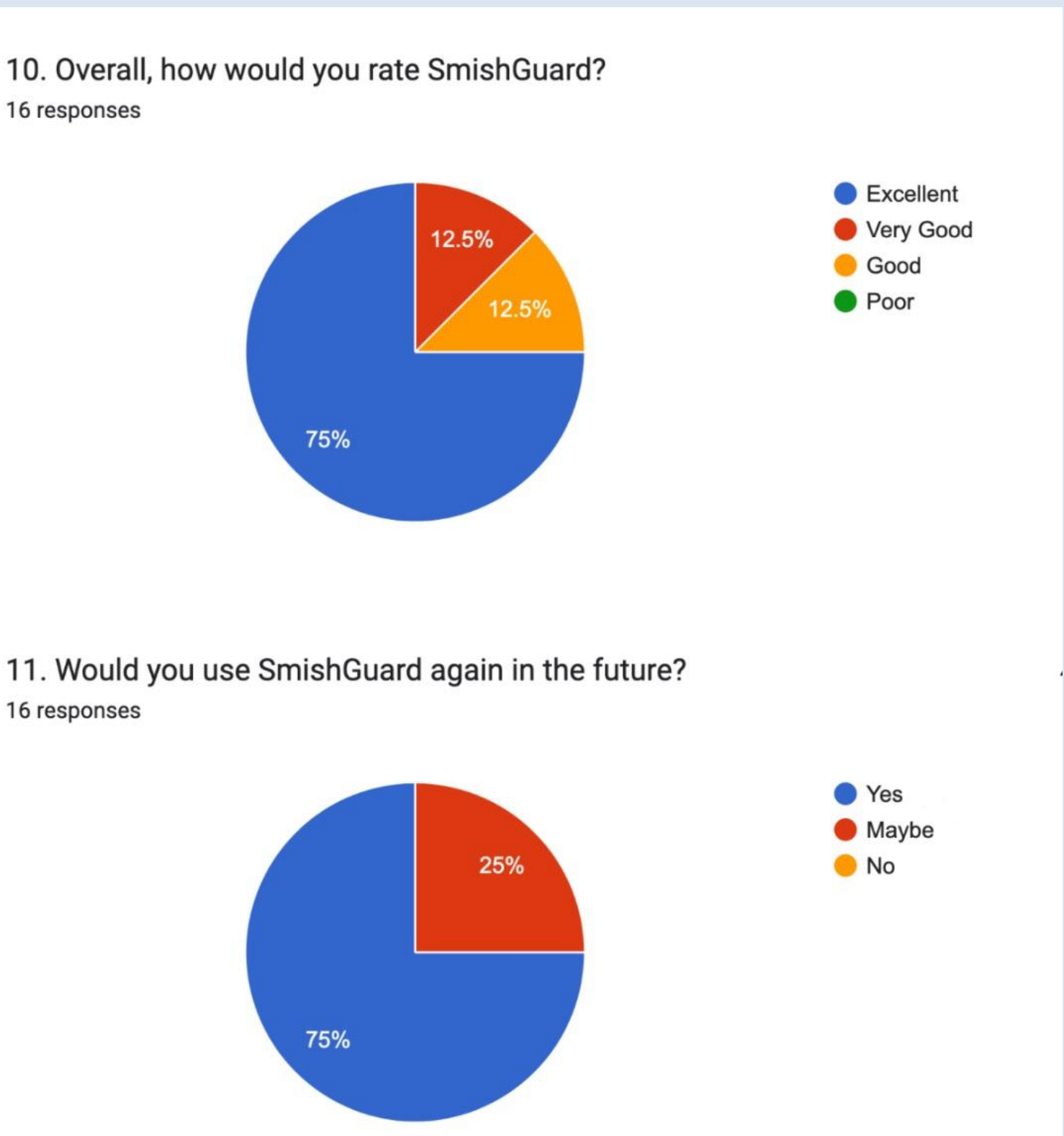


Figure 3: Overall user satisfaction and willingness to reuse the SmishGuard system.

SmishGuard earns a positive rating, with 75% of users ranking it as 'Excellent' and 75% committed to using the platform again, as shown in figure 3.

Conclusion

SmishGuard achieved high performance metrics in smishing detection, outperforming all compared models while providing transparent, educational explanations.

Key Success:

- Superior detection with minimal false positives
- Transparent AI through LIME visualizations
- Bilingual support (English and Arabic)
- Real-time protection with user education

Impact:

SmishGuard bridges the gap between technical AI solutions and user understanding, creating a practical tool that both detects threats and builds security awareness.

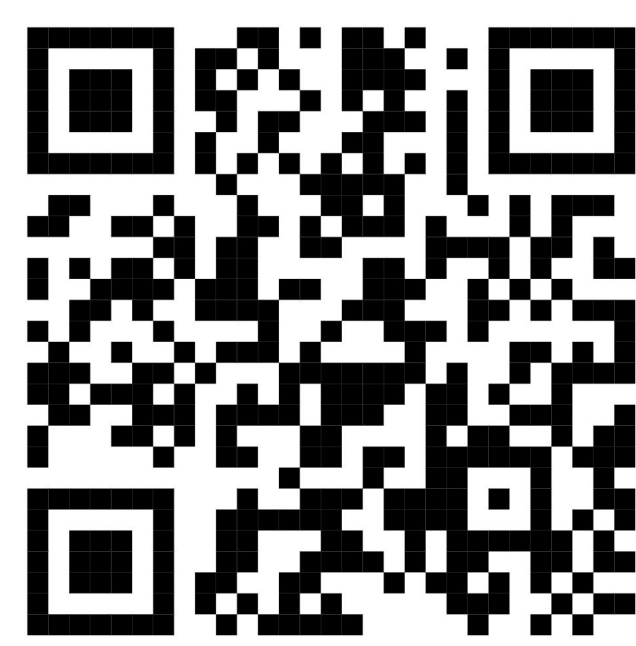
Future Work

- Continuous model retraining with new threats
- Ensemble architecture combining multiple transformers (DistilBERT, BERT, RoBERTa)
- Multi-message SMS conversation analysis
- Native mobile applications (iOS/Android) that analyze SMS messages through real-time smishing alerts.
- Improved Linguistic and Cultural Support
- Cloud-based scalable deployment (AWS/Azure)

References



GitHub



Website Demo:

(Input): This is AI Analysis page where users paste SMS message. It features "quick examples" of common scams (Bank, Delivery) to help users understand typical threats.

 Enter SMS Message

Your delivery has been stopped at our DHL depot. Trk#: R690382803147 Please resolve the issue here: <https://warning-pages.entermypassword.com/join.html?id=>

Analyzing...

Try quick examples:



Bank Alert



Package Delivery



Account Verification

(Results): Provides an immediate, color-coded verdict using explainable AI (LIME). It uses **VirusTotal** integration to cross-reference links against global security databases for instant verification.

 **Analysis Results**

 **Classification**

 **SMISHING DETECTED**

 **Detection Method: VirusTotal URL Analysis**

This message contains URLs flagged as harmful by VirusTotal security engines. The AI text analysis was not used for this classification.

 **Harmful URLs Found:** 1

 **Malicious Flags:** 1 security engine(s)

 **Suspicious Flags:** 0 security engine(s)



HARMFUL URLS

1



MALICIOUS FLAGS

1



DETECTION METHOD

VirusTotal

(Breakdown): Shows the technical "why." It highlights the specific URL flagged as harmful, explaining that the link itself is the primary evidence of the scam.

Analysis Breakdown

 This message was classified based on URL analysis only. Text-based AI analysis was not performed because VirusTotal detected harmful URLs.

VirusTotal URL Scan Results

URLs Found: 1

URL 1:

<https://warning-pages.entermypassword.com/join.html?id>

 **Status: HARMFUL**

(Safety): Lists critical "Do's and Don'ts." It instructs the user to delete the message and includes a disclaimer that the tool is for educational guidance.

Safety Recommendations

 **CRITICAL: VirusTotal detected malicious URLs!**

 **Do not engage** with this message

 **Do not click** any links in the message

 **Verify directly** with the organization using official contact methods

 **Delete** the message immediately

 **Stay vigilant:** Always verify unexpected messages through official channels

  **Important Disclaimer:** SmishGuard is an educational tool designed to help identify potential smishing threats. While we use advanced AI and security databases, no automated system is 100% accurate. Always exercise caution with suspicious messages and verify through official channels. SmishGuard and its creators are not responsible for any decisions made based on these results.

(Action Check): A diagnostic screen that asks the user how they interacted with the message (e.g., "Clicked Link") to determine the level of risk.

What Action Did You Take?

Select the action you took after receiving this message to get specific guidance



Clicked on Link



Revealed Credentials



Shared Financial Info



Downloaded/Installed



Shared Personal Info



Replied to Message



Made a Payment



Scanned QR Code



Called a Number



No Action Yet

(Mitigation): Offers a tailored recovery plan. It provides immediate steps to secure the device and long-term tips to prevent future successful attacks.

Clicked on Link

Immediate Actions

- ▶ Close the webpage immediately without entering any information
- ▶ Do not download anything from the site
- ▶ Clear your browser cache and cookies
- ▶ Run a security scan on your device using antivirus software

Prevention Tips

- ▶ Never click links in unsolicited messages
- ▶ Verify sender identity through official channels first
- ▶ Hover over links (on desktop) to preview the actual URL
- ▶ Look for suspicious URLs (misspellings, unusual domains, random characters)

Another examples:

A Smish message without URL

Enter SMS Message

Congratulations! You've won a cash prize. Provide your ID number to claim.

Analysis Results

Classification

SMISHING DETECTED

AI Model: Fine-tuned DistilBERT

Confidence: 99.9% (VERY HIGH)

Probability Breakdown: Legitimate: 0.1% | Smishing: 99.9%

AI MODEL

Fine-tuned DistilBERT

CONFIDENCE

99.9%

URLS FOUND

0

MESSAGE ANALYSIS:

Congratulations ! You've won a cash prize . Provide your ID number to claim .

Final Verdict: The AI identified key threat indicators and classified this as SMISHING

A Legitimate Message

Enter SMS Message

Reminder: Your Google Drive storage is almost full. Manage your files at https://drive.google.com

Analysis Results

Classification

APPEARS LEGITIMATE

AI Model: Fine-tuned DistilBERT

Confidence: 99.6% (VERY HIGH)

Probability Breakdown: Legitimate: 99.6% | Smishing: 0.4%

AI MODEL

Fine-tuned DistilBERT

CONFIDENCE

99.6%

URLS FOUND

1

MESSAGE ANALYSIS:

Reminder: Your Google Drive storage is almost full . Manage your files at https : //drive . google . com

Final Verdict: The AI identified key legitimate patterns and classified this as HAM